

XICHANG, China

WMO: 565710

Lat: 27.9035N

Lon: 102.2678E

Elev: 1599

StdP: 83.53

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.0	3.3	-10.1	1.9	18.3	-8.1	2.3	16.5	6.7	13.7	5.8	14.8	0.7	0	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	32.6	17.9	31.4	18.0	30.3	18.1	21.6	27.4	21.1	26.7	20.6	26.0	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	18.1	23.0	19.7	17.5	22.8	19.2	17.0	22.7	71.5	27.3	69.6	26.8	67.9	26.1	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.3	4.4	DB	-0.2	34.8	1.2	1.7	-1.1	36.0	-1.8	37.0	-2.5	37.9	-3.4	39.1	(4)
(5)			WB	-2.1	22.6	1.3	0.9	-3.0	23.3	-3.8	23.8	-4.6	24.3	-5.5	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	11.0	13.5	17.0	20.3	22.2	22.4	23.5	23.5	21.1	18.2	14.5	11.1	(6)	
	DBStd	5.51	3.02	4.13	3.68	3.80	3.91	3.15	2.39	2.36	3.13	2.95	2.70	2.51	(7)	
	HDD10.0	63	23	16	3	0	0	0	0	0	0	0	3	18	(8)	
	HDD18.3	866	227	139	67	25	13	4	1	1	11	38	116	226	(9)	
	CDD10.0	3058	54	114	220	308	377	372	418	418	332	254	139	51	(10)	
	CDD18.3	819	0	4	26	82	132	126	160	160	93	34	2	0	(11)	
	CDH23.3	5782	1	38	271	794	1166	872	974	987	507	162	11	0	(12)	
(13)	CDH26.7	1779	0	1	36	253	473	293	273	303	132	15	0	0	(13)	
(14)	Wind	WSAvg	1.4	1.5	1.9	2.1	2.0	1.7	1.1	1.1	1.1	1.1	1.2	1.2	(14)	
(15) Precipitation	PrecAvg	971	5	5	11	30	89	209	217	177	159	74	18	6	(15)	
	PrecMax	1473	24	29	45	89	214	379	539	412	282	191	82	50	(16)	
	PrecMin	557	0	0	0	3	12	108	83	50	37	9	0	0	(17)	
	PrecStd	163	6	6	10	20	49	63	94	79	57	40	17	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	26.3	29.2	32.6	34.8	34.2	32.4	32.7	31.8	28.3	24.7	21.5	(19)	
		MCWB	9.3	10.5	12.5	14.8	16.8	19.3	20.6	19.7	19.6	17.7	13.7	9.9	(20)	
	2%	DB	20.6	24.5	27.6	31.0	33.0	31.6	30.9	31.2	29.8	26.7	23.1	19.8	(21)	
		MCWB	8.6	9.9	12.0	14.2	16.5	19.3	20.4	19.9	19.4	17.0	12.9	9.7	(22)	
	5%	DB	19.2	23.0	26.2	29.6	31.3	30.0	29.6	29.9	28.1	25.2	21.8	18.5	(23)	
		MCWB	8.1	9.5	11.5	13.9	16.5	19.3	20.3	19.8	18.8	16.3	12.5	9.4	(24)	
	10%	DB	17.8	21.4	24.8	28.0	29.5	28.1	28.1	28.4	26.4	23.6	20.3	17.0	(25)	
		MCWB	7.8	8.9	11.0	13.6	16.3	19.1	20.2	19.7	18.5	15.9	12.1	9.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	12.0	14.1	17.1	19.8	21.7	22.2	22.2	20.9	18.9	15.7	11.7	(27)	
		MCDB	19.3	23.5	25.1	28.3	29.0	27.8	28.0	28.8	27.0	24.7	21.5	19.1	(28)	
	2%	WB	9.6	10.8	13.1	15.9	18.9	21.1	21.6	21.4	20.2	18.1	14.5	10.6	(29)	
		MCDB	18.5	21.9	24.5	26.4	27.3	26.9	27.2	27.4	26.4	24.5	20.5	18.0	(30)	
	5%	WB	8.9	10.1	12.4	15.1	18.2	20.6	21.2	20.8	19.6	17.4	13.5	9.8	(31)	
		MCDB	17.8	21.3	23.8	25.3	26.2	26.1	26.6	26.5	25.6	23.3	19.8	17.1	(32)	
	10%	WB	8.1	9.4	11.7	14.4	17.5	20.1	20.7	20.4	19.1	16.7	12.7	9.2	(33)	
		MCDB	16.7	20.2	22.9	24.8	25.5	25.3	25.9	25.7	24.6	22.0	18.8	16.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.2	12.7	12.2	10.5	8.1	7.9	8.5	7.8	8.1	9.8	10.4	(35)	
		MCDBR	13.6	14.4	14.8	14.4	13.6	11.5	10.4	11.0	10.7	10.9	12.4	13.0	(36)	
	5% WB	MCWBR	5.7	5.8	5.4	4.3	3.7	3.1	3.0	3.2	3.1	3.5	4.6	5.5	(37)	
		MCWBR	13.3	13.9	14.0	13.3	11.2	9.0	8.6	8.9	9.0	9.3	10.7	12.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.332	0.349	0.462	0.488	0.472	0.466	0.474	0.489	0.471	0.435	0.374	0.345	(40)	
		taud	2.369	2.339	1.993	1.984	2.116	2.206	2.181	2.116	2.141	2.182	2.288	2.343	(41)	
	Ebn at Noon	Ebn at Noon	909	921	829	819	830	830	822	808	811	818	851	873	(42)	
		Edh at Noon	104	117	175	182	160	145	149	158	149	135	113	103	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.75	4.62	5.12	5.41	5.19	4.23	4.32	4.21	3.56	3.34	3.54	3.30	(44)	
	RadStd	RadStd	0.37	0.51	0.46	0.59	0.44	0.58	0.37	0.53	0.38	0.39	0.45	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.41	N/A	-1.52	+1.10	-0.20	N/A	N/A	-92	+190	+114
(47)	Regional (4 neighbors)	+0.47	N/A	N/A	+0.94	N/A	N/A	N/A	-92	+190	+114

Nomenclature: See separate page